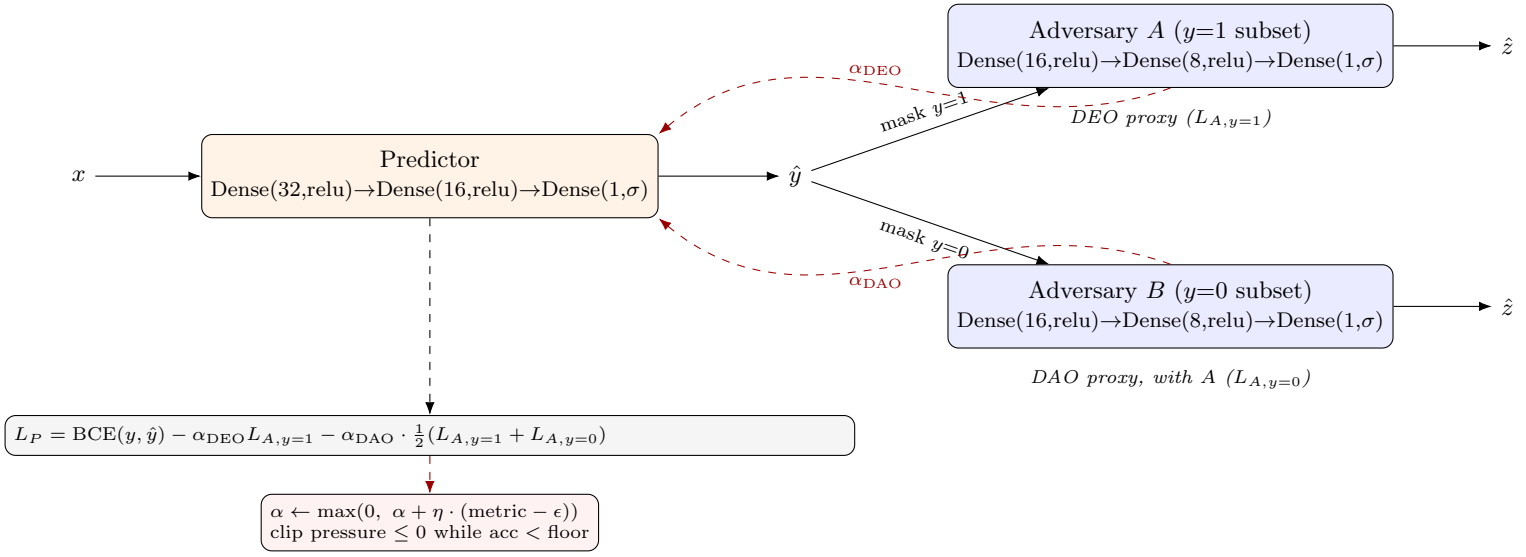

Adversary A (y=1 subset)
Dense(16,relu)→Dense(8,relu)→Dense(1,σ)
$\hat{z}$
α_{DEO}
mask y=1
DEO proxy ($L_{A,y=1}$)
x
Predictor
Dense(32,relu)→Dense(16,relu)→Dense(1,σ)
$\hat{y}$
mask y=0
α_{DAO}
Adversary B (y=0 subset)
Dense(16,relu)→Dense(8,relu)→Dense(1,σ)
$\hat{z}$
DAO proxy, with A ($L_{A,y=0}$)
$L_P = \text{BCE}(y,\hat{y}) - \alpha_{\text{DEO}} L_{A,y=1} - \alpha_{\text{DAO}} \cdot \frac{1}{2}(L_{A,y=1} + L_{A,y=0})$
$\alpha \leftarrow \max(0,\ \alpha + \eta \cdot (\text{metric} - \epsilon))$
clip pressure ≤ 0 while acc $<$ floor